

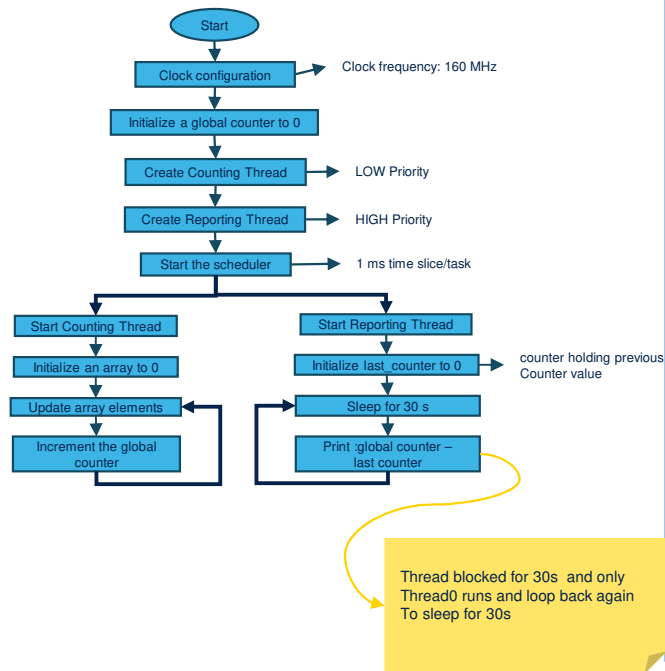
FreeRTOS v11.2



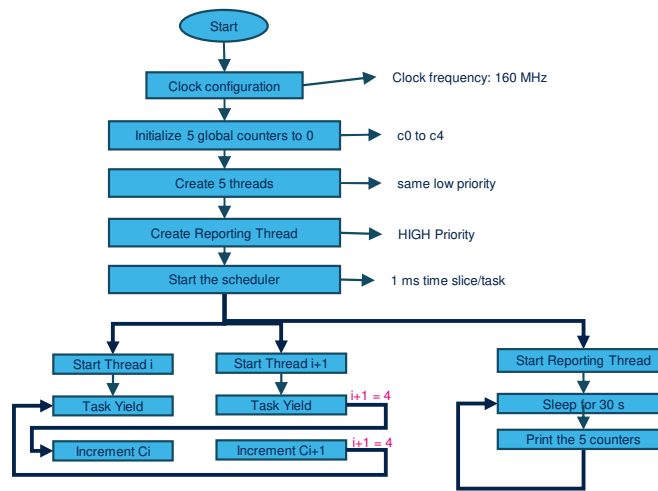
STM32 vs GD32 MW Benchmark: FreeRTOS

Flowcharts

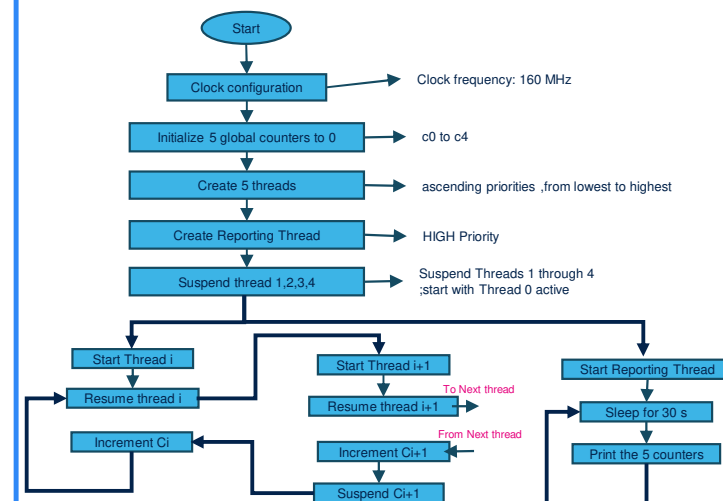
Basic Processing Flowchart



Cooperative Processing Flowchart



Preemptive Scheduling Flowchart



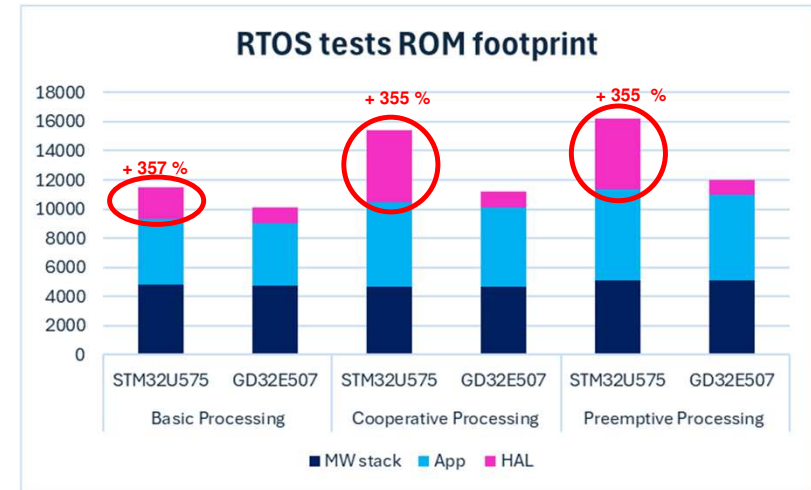
STM32 vs GD32 MW Benchmark: FreeRTOS Tests

FreeRTOS tests Footprint

RTOS Scenario	Footprint	STM32U575	GD32E507	STM232 vs GD32
Basic Processing	ROM	14,266	10,128	+ 40.86 %
	RAM	11,960	11,716	+2.08 %
Cooperative Processing	ROM	15,422	11,224	+ 37 %
	RAM	10,972	10,728	+2.27 %
Preemptive Processing	ROM	16,226	12,028	+ 35 %
	RAM	10,996	10,748	+ 2.31 %

FreeRTOS tests ROM Footprint

RTOS Scenario	Footprint	STM32U575	GD32E507	STM232 vs GD32
Basic Processing ROM	App	4531	4278	+ 6 %
	HAL	4927	1078	+ 357 %
	MW stack	4808	4772	+ 0.7 %
Cooperative Processing ROM	App	5823	5484	+ 6 %
	HAL	4911	1078	+ 355 %
	MW stack	4688	4662	+ 0.55 %
Preemptive Processing ROM	App	6191	5810	+ 6 %
	HAL	4911	1078	+ 355 %
	MW stack	5124	5140	-0.3 %



FreeRTOS tests performance (thread/s)

RTOS Scenario	STM32U575	GD32E507	STM232 vs GD32
Basic Processing (Thread/s)	47,943	47,955	- 0.025 %
Cooperative Processing (Thread/30s)	1,220,066	1,200,666	+ 1.6 %
Preemptive Processing (Thread/30s)	294,615	285,144	+ 3.3 %

$$STM32 \text{ vs } GD32 (\%) = (STM32 - GD32) / GD32$$

- STM32 ROM footprint is **higher by average 37%** than GD32
- Although **tests have almost the same Application and FreeRTOS stack footprint**, but there is a big gap in HAL drivers' footprint (mainly due to HAL RCC footprint)
- FreeRTOS test have almost the same performance

